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Aboulhosn

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(54) **FEE RE-RATING FOR A WIRELESS CARRIER NETWORK**

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(51) **Int. Cl.**

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H04M 15/00 (2024.01)
H04M 15/10 (2006.01)
H04W 8/18 (2009.01)

(52) **U.S. Cl.**

CPC ***H04W 4/24*** (2013.01); ***H04M 15/10*** (2013.01); ***H04M 15/41*** (2013.01); ***H04M 15/80*** (2013.01); ***H04W 8/18*** (2013.01)

(58) **Field of Classification Search**

CPC H04W 4/24; H04W 8/18; H04M 15/10; H04M 15/41; H04M 15/80; H04M 15/00; H04M 15/43; H04M 15/44; H04M 15/58; H04M 15/70; H04M 15/73

See application file for complete search history.

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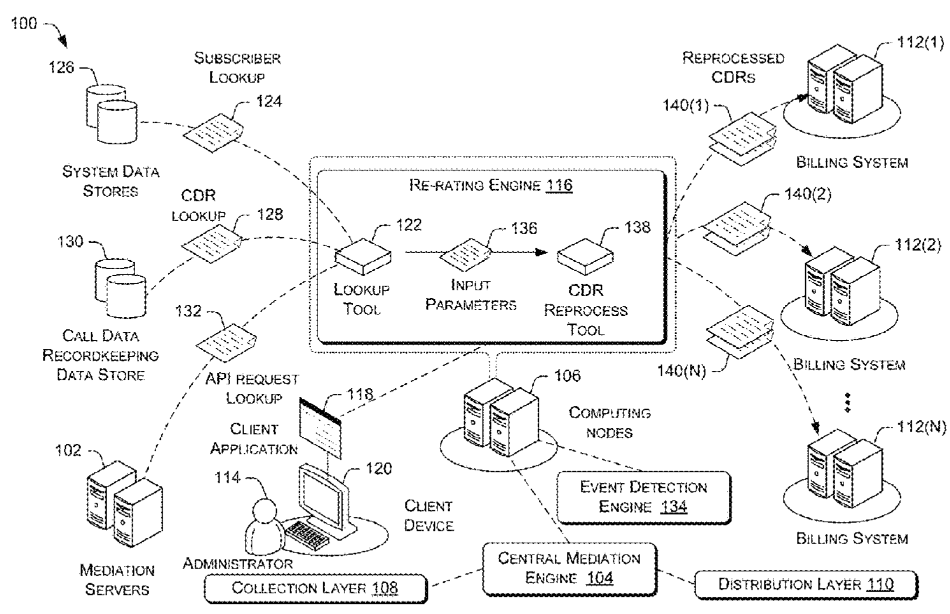
Primary Examiner — Matthew D. Anderson
Assistant Examiner — Abdullah Al Mamun

(57) **ABSTRACT**

A business logic used to generate call details records (CDRs) for a plurality of subscribers by a wireless carrier network is imported. A selection of one or more data collector applications that are listed in the business logic is received and one or more configuration parameters in the business logic are updated based on one or more inputted parameter updates. A selection of one or more CDR distributor applications listed in the business logic is also received. A particular set of CDRs in the specific group of CDRs is further selected based on one or more inputted selection parameters. Reprocessed CDRs that correspond to the particular set of CDRs are generated according to at least the one or more updated configuration parameters in the business logic and sent to the one or more billing systems via the one or more CDR distributor applications to trigger recalculation of service fees.

20 Claims, 11 Drawing Sheets



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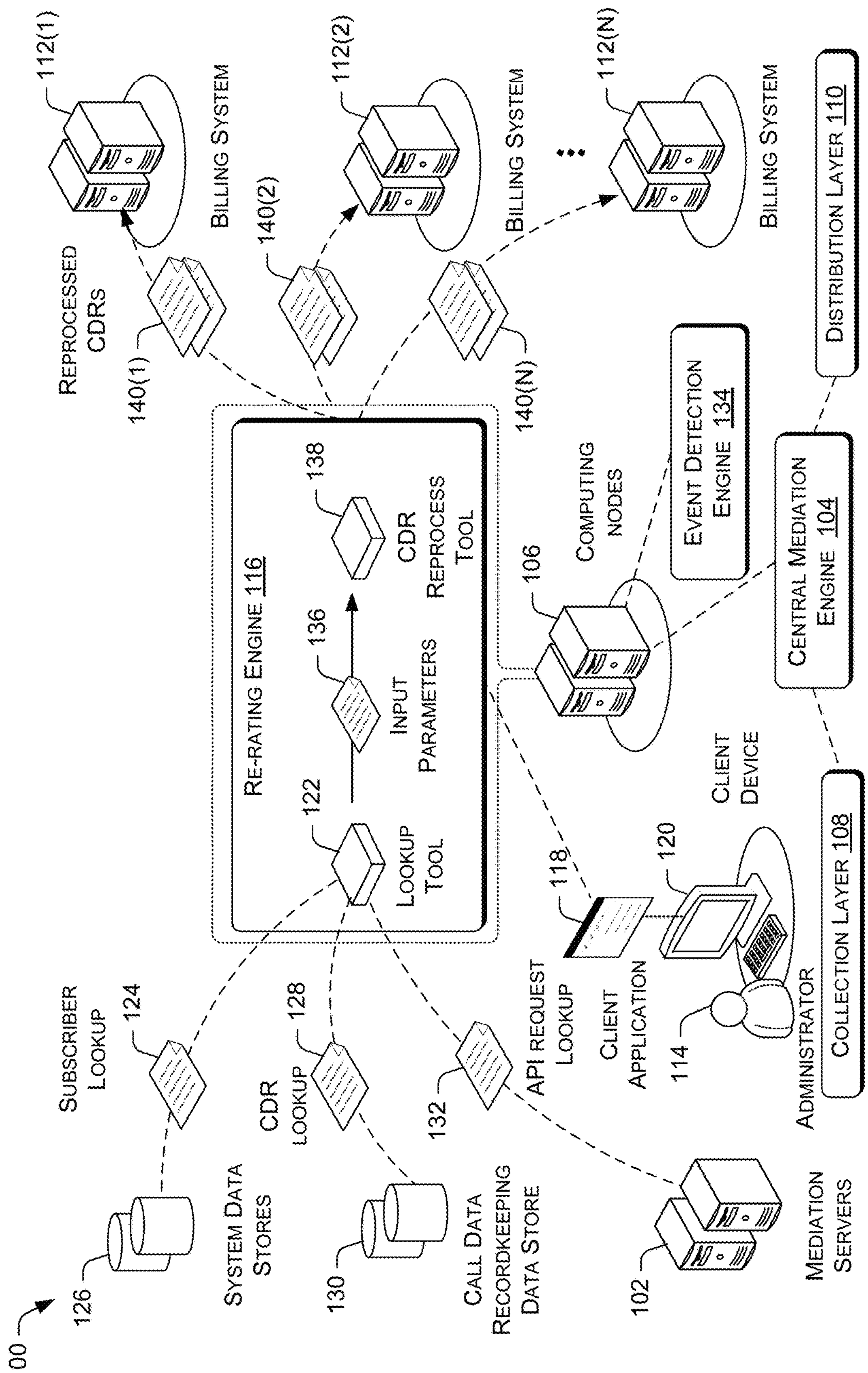
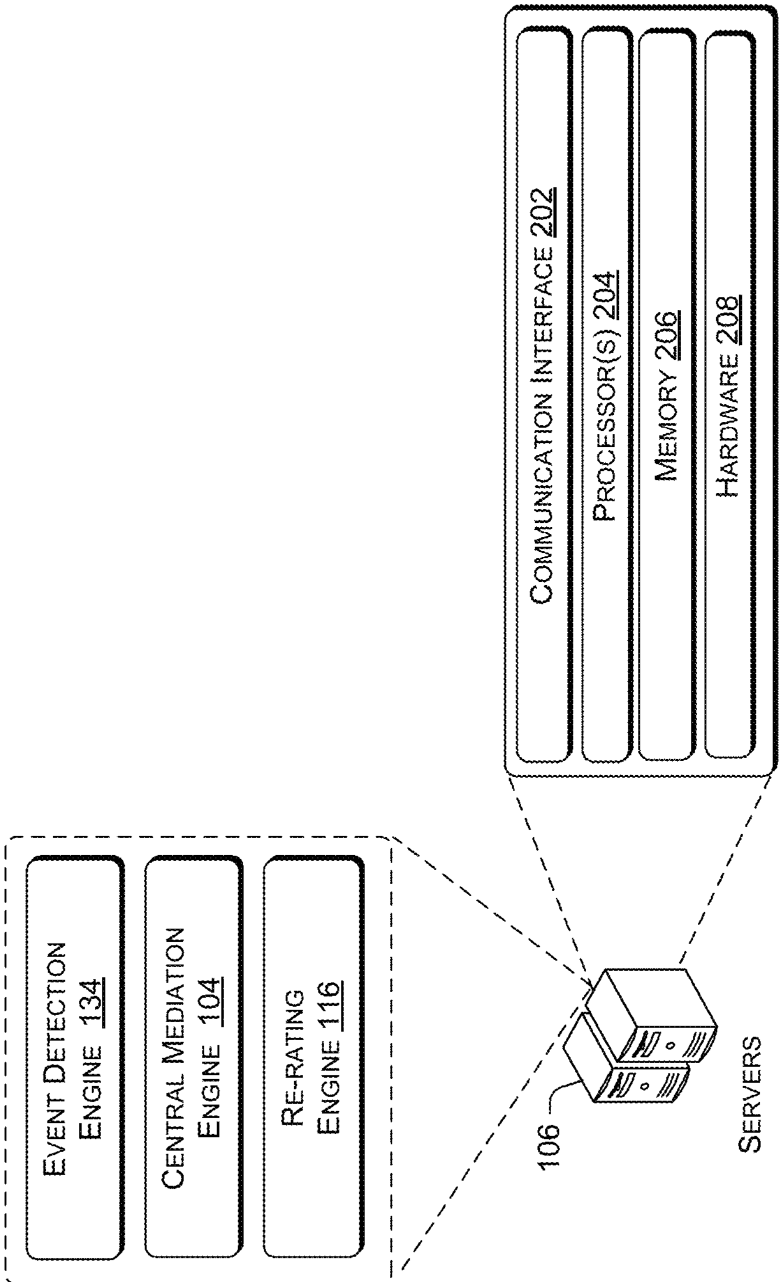

100
126
System Data Stores
124
Subscriber Lookup
130
Call Data Recordkeeping Data Store
128
CDR Lookup
102
Mediation Servers
132
API request Lookup
Re-Rating Engine 116
122
Lookup Tool
136
Input Parameters
138
CDR Reprocess Tool
118
Client Application
120
Client Device
114
Administrator
106
Computing Nodes
Event Detection Engine 134
Central Mediation Engine 104
Collection Layer 108
Distribution Layer 110
Reprocessed CDRs
140(1)
140(2)
140(N)
112(1)
112(2)
112(N)
Billing System
Billing System
Billing System

FIG. 1



EVENT DETECTION ENGINE 134
CENTRAL MEDIATION ENGINE 104
RE-RATING ENGINE 116
106
SERVERS
COMMUNICATION INTERFACE 202
PROCESSOR(S) 204
MEMORY 206
HARDWARE 208

FIG. 2

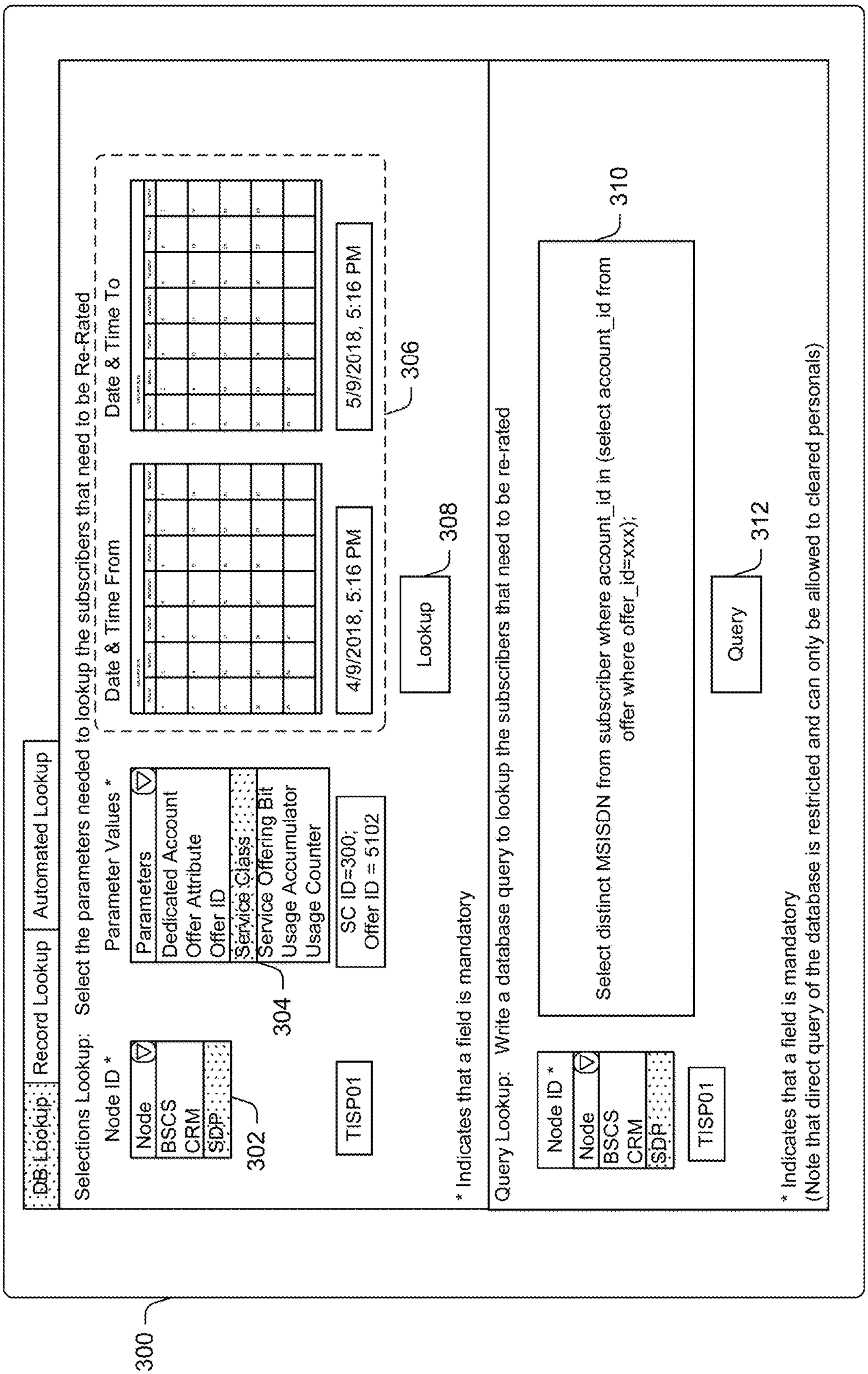

DB Lookup
Record Lookup
Automated Lookup
Selections Lookup: Select the parameters needed to lookup the subscribers that need to be Re-Rated
Node ID *
Node
BSCS
CRM
SDP
TISP01
302
Parameter Values *
Parameters
Dedicated Account
Offer Attribute
Offer ID
Service Class
Service Offering Bit
Usage Accumulator
Usage Counter
SC ID=300;
Offer ID = 5102
304
Date & Time From
4/9/2018, 5:16 PM
Date & Time To
5/9/2018, 5:16 PM
306
Lookup
308
* Indicates that a field is mandatory
Query Lookup: Write a database query to lookup the subscribers that need to be re-rated
Node ID *
Node
BSCS
CRM
SDP
TISP01
Select distinct MSISDN from subscriber where account_id in (select account_id from offer where offer_id=xxx);
310
Query
312
* Indicates that a field is mandatory
(Note that direct query of the database is restricted and can only be allowed to cleared personals)
300

FIG. 3a

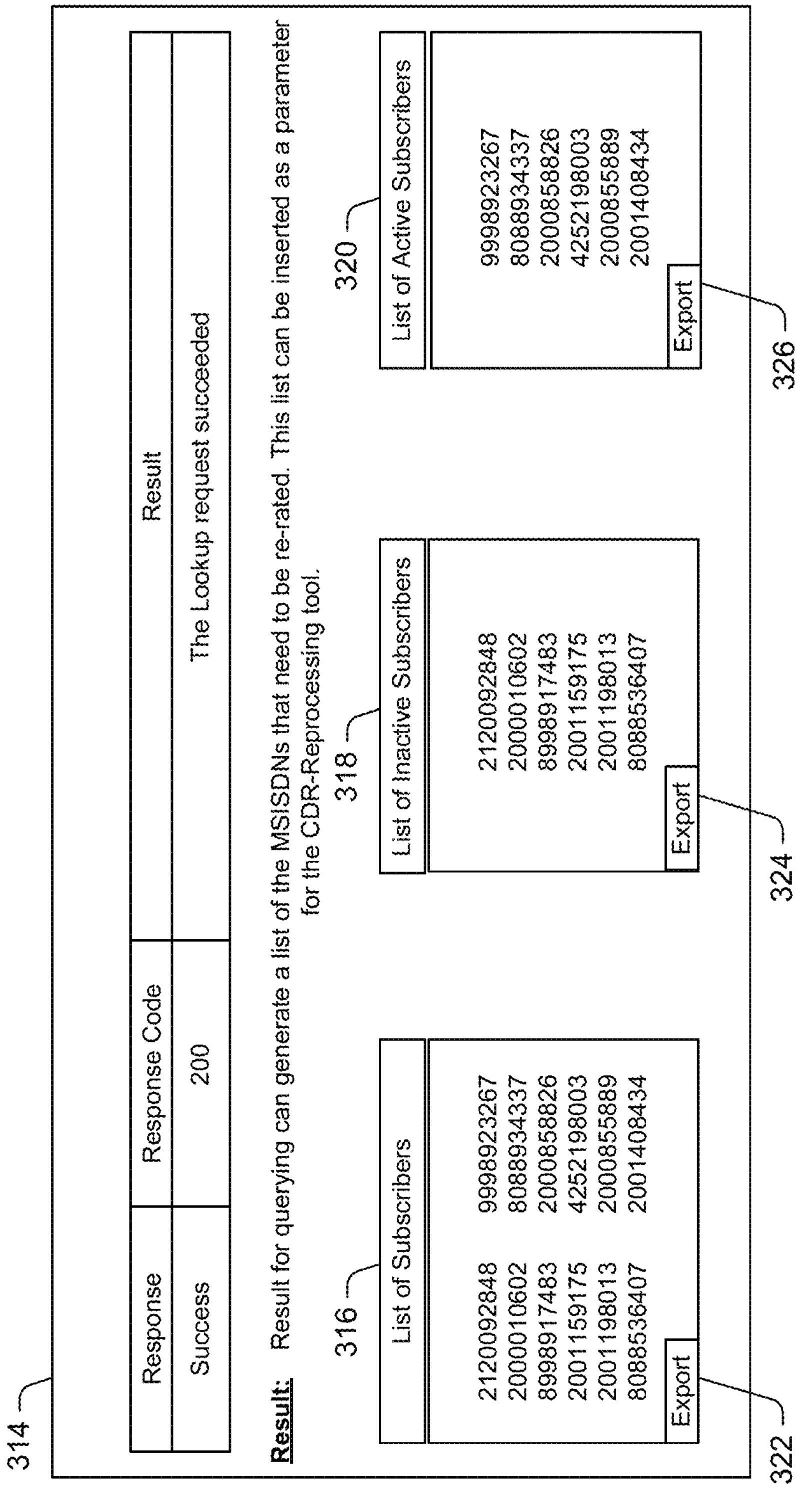

314
Response
Success
Response Code
200
Result
The Lookup request succeeded
Result: Result for querying can generate a list of the MSISDNs that need to be re-rated. This list can be inserted as a parameter for the CDR-Reprocessing tool.
316
List of Subscribers
2120092848
2000010602
8998917483
2001159175
2001198013
8088536407
9998923267
8088934337
2000858826
4252198003
2000855889
2001408434
Export
322
318
List of Inactive Subscribers
2120092848
2000010602
8998917483
2001159175
2001198013
8088536407
Export
324
320
List of Active Subscribers
9998923267
8088934337
2000858826
4252198003
2000855889
2001408434
Export
326

FIG. 3b

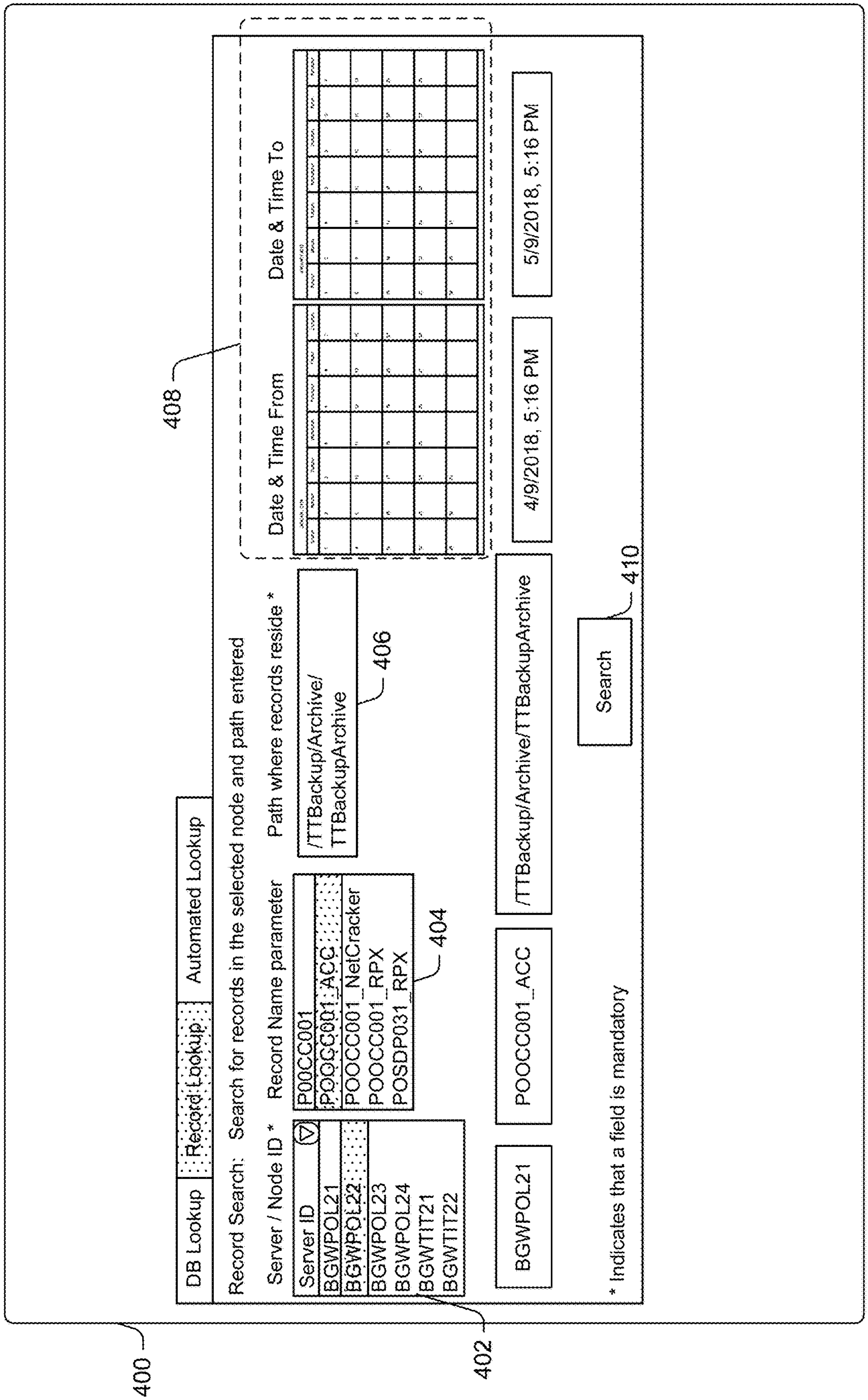

DB Lookup
Record Lookup
Automated Lookup
Record Search: Search for records in the selected node and path entered
Server / Node ID *
Record Name parameter
Path where records reside *
Date & Time From
Date & Time To
Server ID
BGWPOL21
BGWPOL22
BGWPOL23
BGWPOL24
BGWTIT21
BGWTIT22
P0OCC001
POOCC001_ACC
POOCC001_NetCracker
POOCC001_RPX
POSDP031_RPX
/TTBackup/Archive/
TTBackupArchive
BGWPOL21
POOCC001_ACC
/TTBackup/Archive/TTBackupArchive
4/9/2018, 5:16 PM
5/9/2018, 5:16 PM
Search
* Indicates that a field is mandatory
400
402
404
406
408
410

FIG. 4a

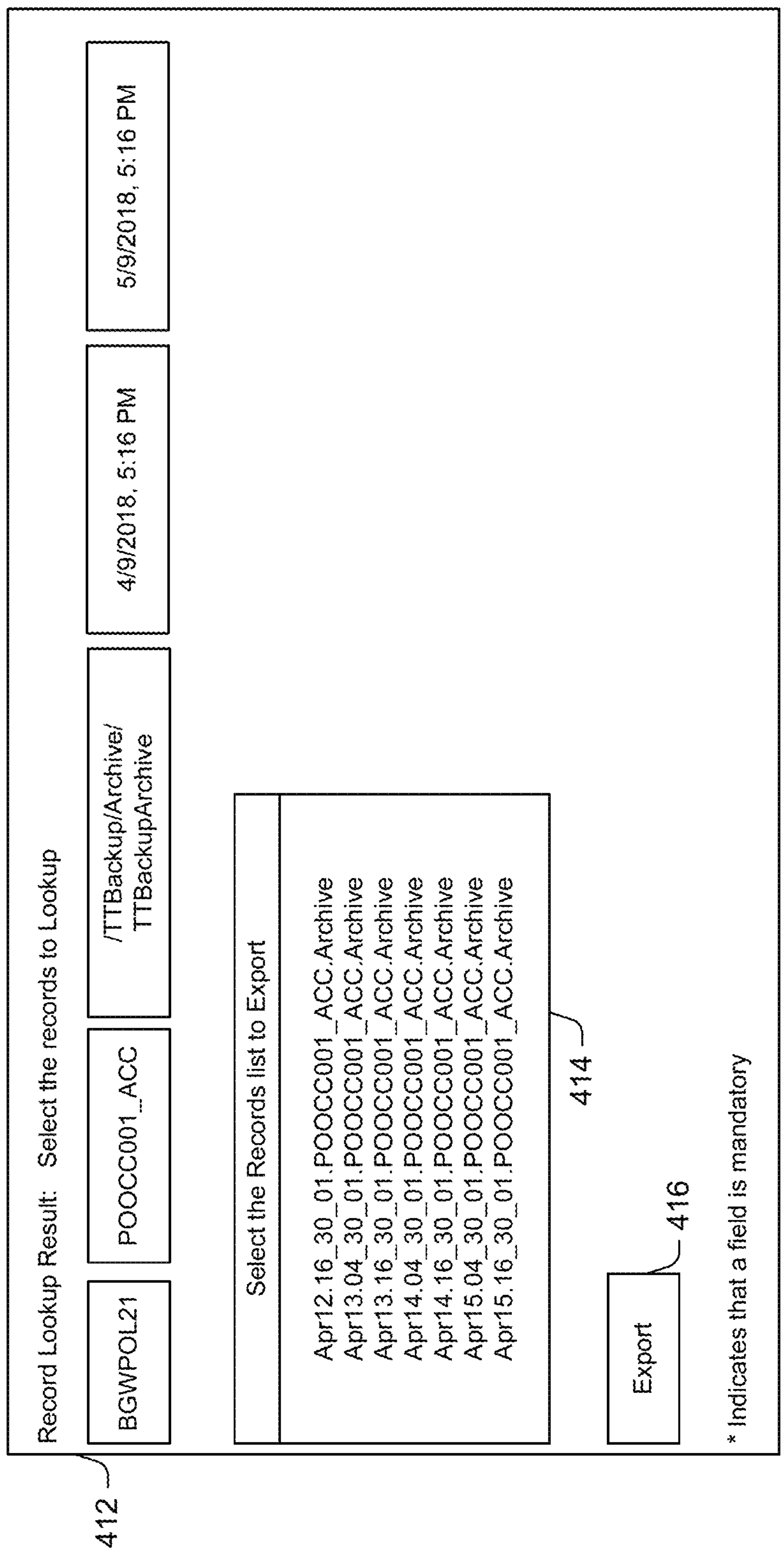

Record Lookup Result: Select the records to Lookup
BGWPOL21
POOCC001_ACC
/TTBackup/Archive/ TTBackupArchive
4/9/2018, 5:16 PM
5/9/2018, 5:16 PM
Select the Records list to Export
Apr12.16_30_01.POOCC001_ACC.Archive
Apr13.04_30_01.POOCC001_ACC.Archive
Apr13.16_30_01.POOCC001_ACC.Archive
Apr14.04_30_01.POOCC001_ACC.Archive
Apr14.16_30_01.POOCC001_ACC.Archive
Apr15.04_30_01.POOCC001_ACC.Archive
Apr15.16_30_01.POOCC001_ACC.Archive
414
Export
416
* Indicates that a field is mandatory
412

FIG. 4b

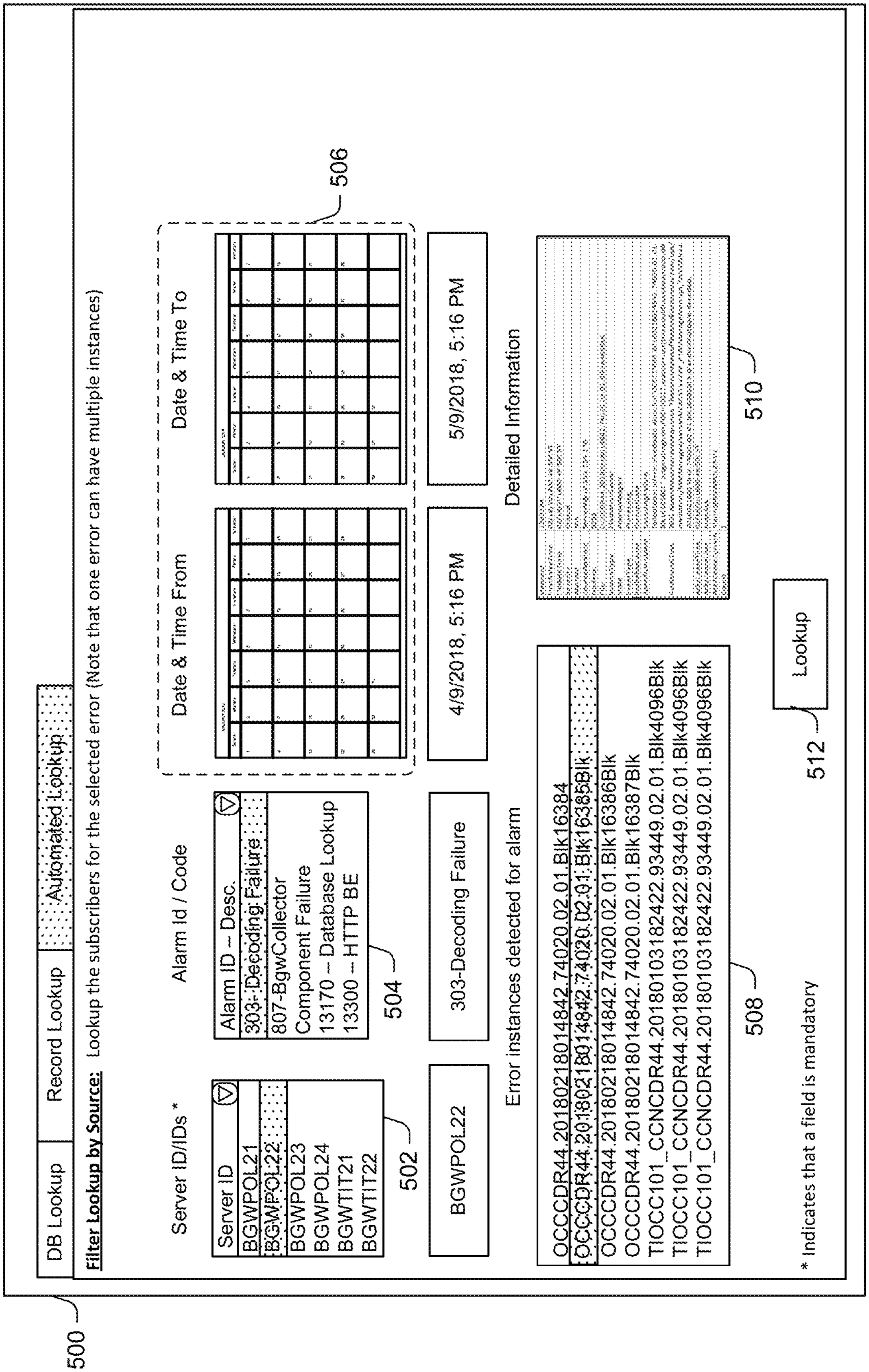

DB Lookup
Record Lookup
Automated Lookup
Filter Lookup by Source: Lookup the subscribers for the selected error (Note that one error can have multiple instances)
Server ID/IDs *
Server ID
BGWPOL21
BGWPOL22
BGWPOL23
BGWPOL24
BGWTIT21
BGWTIT22
502
BGWPOL22
Alarm Id / Code
Alarm ID -- Desc.
303-Decoding Failure
807-BgwCollector Component Failure
13170 -- Database Lookup
13300 -- HTTP BE
504
303-Decoding Failure
Date & Time From
4/9/2018, 5:16 PM
Date & Time To
5/9/2018, 5:16 PM
506
Error instances detected for alarm
OCCCDR44.20180218014842.74020.02.01.Blk16384
OCCCDR44.20180218014842.74020.02.01.Blk16385Blk
OCCCDR44.20180218014842.74020.02.01.Blk16386Blk
OCCCDR44.20180218014842.74020.02.01.Blk16387Blk
TIOCC101_CCNCDR44.20180103182422.93449.02.01.Blk4096Blk
TIOCC101_CCNCDR44.20180103182422.93449.02.01.Blk4096Blk
TIOCC101_CCNCDR44.20180103182422.93449.02.01.Blk4096Blk
508
Detailed Information
510
Lookup
512
* Indicates that a field is mandatory
500

FIG. 5a

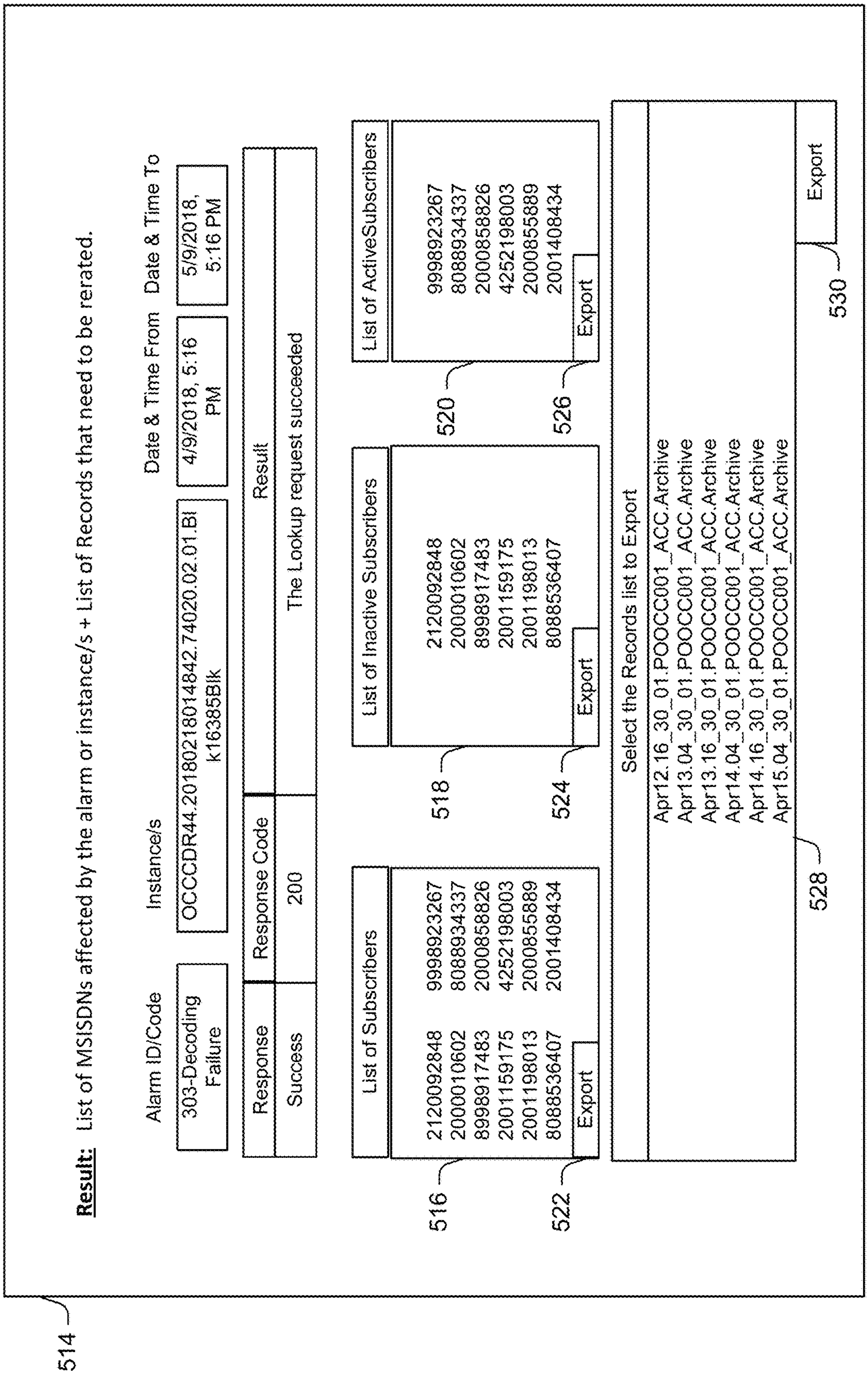

514
Result: List of MSISDNs affected by the alarm or instance/s + List of Records that need to be rerated.
Alarm ID/Code
303-Decoding Failure
Instance/s
OCCCDR44.20180218014842.74020.02.01.Blk16385Blk
Date & Time From
4/9/2018, 5:16 PM
Date & Time To
5/9/2018, 5:16 PM
Response
Success
Response Code
200
Result
The Lookup request succeeded
516
List of Subscribers
2120092848 9998923267
2000010602 8088934337
8998917483 2000858826
2001159175 4252198003
2001198013 2000855889
8088536407 2001408434
Export
522
518
List of Inactive Subscribers
2120092848
2000010602
8998917483
2001159175
2001198013
8088536407
Export
524
520
List of ActiveSubscribers
9998923267
8088934337
2000858826
4252198003
2000855889
2001408434
Export
526
Select the Records list to Export
Apr12.16_30_01.POOCC001_ACC.Archive
Apr13.04_30_01.POOCC001_ACC.Archive
Apr13.16_30_01.POOCC001_ACC.Archive
Apr14.04_30_01.POOCC001_ACC.Archive
Apr14.16_30_01.POOCC001_ACC.Archive
Apr15.04_30_01.POOCC001_ACC.Archive
528
Export
530

FIG. 5b

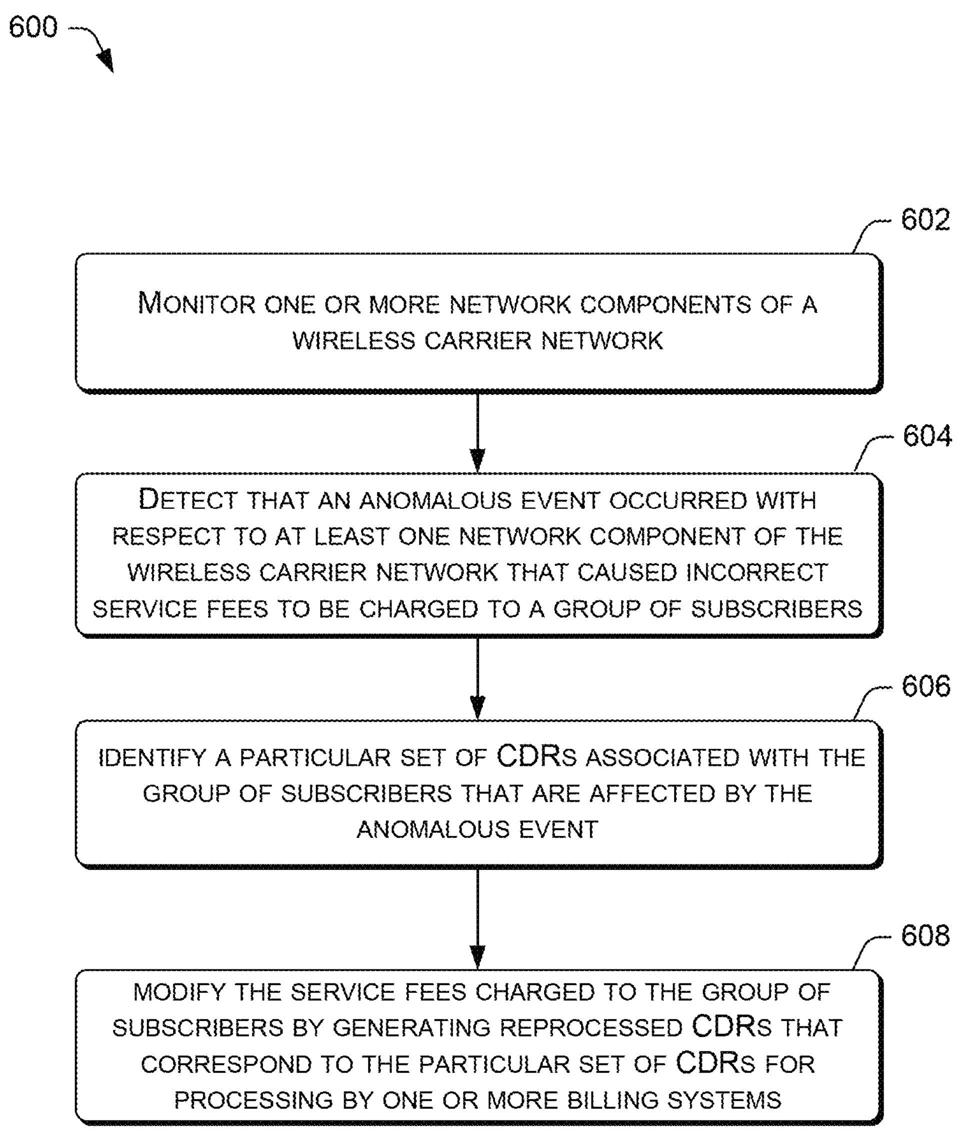

600
602
MONITOR ONE OR MORE NETWORK COMPONENTS OF A WIRELESS CARRIER NETWORK
604
DETECT THAT AN ANOMALOUS EVENT OCCURRED WITH RESPECT TO AT LEAST ONE NETWORK COMPONENT OF THE WIRELESS CARRIER NETWORK THAT CAUSED INCORRECT SERVICE FEES TO BE CHARGED TO A GROUP OF SUBSCRIBERS
606
IDENTIFY A PARTICULAR SET OF CDRS ASSOCIATED WITH THE GROUP OF SUBSCRIBERS THAT ARE AFFECTED BY THE ANOMALOUS EVENT
608
MODIFY THE SERVICE FEES CHARGED TO THE GROUP OF SUBSCRIBERS BY GENERATING REPROCESSED CDRS THAT CORRESPOND TO THE PARTICULAR SET OF CDRS FOR PROCESSING BY ONE OR MORE BILLING SYSTEMS

FIG. 6

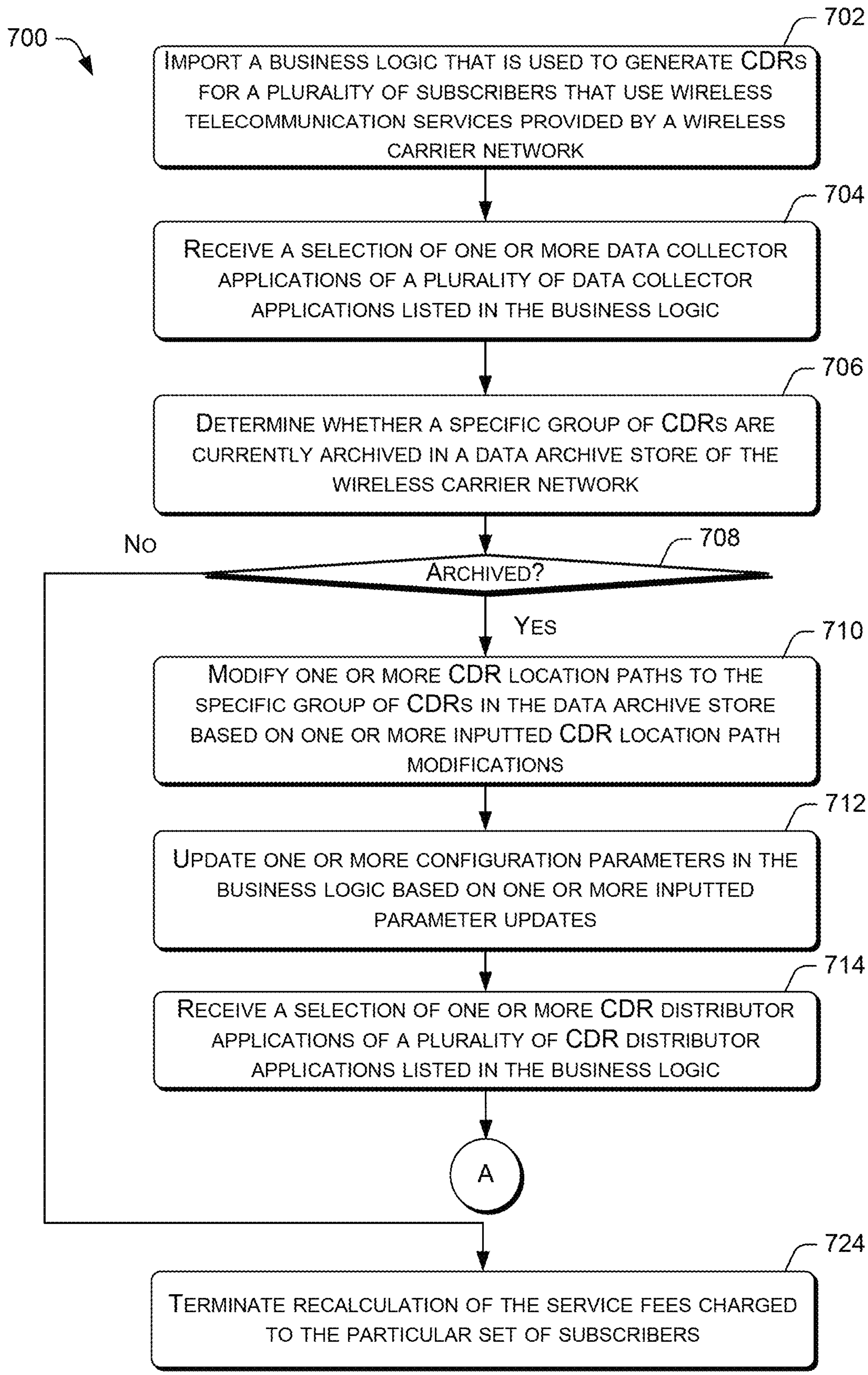

700
702
IMPORT A BUSINESS LOGIC THAT IS USED TO GENERATE CDRS FOR A PLURALITY OF SUBSCRIBERS THAT USE WIRELESS TELECOMMUNICATION SERVICES PROVIDED BY A WIRELESS CARRIER NETWORK
704
RECEIVE A SELECTION OF ONE OR MORE DATA COLLECTOR APPLICATIONS OF A PLURALITY OF DATA COLLECTOR APPLICATIONS LISTED IN THE BUSINESS LOGIC
706
DETERMINE WHETHER A SPECIFIC GROUP OF CDRS ARE CURRENTLY ARCHIVED IN A DATA ARCHIVE STORE OF THE WIRELESS CARRIER NETWORK
708
ARCHIVED?
NO
YES
710
MODIFY ONE OR MORE CDR LOCATION PATHS TO THE SPECIFIC GROUP OF CDRS IN THE DATA ARCHIVE STORE BASED ON ONE OR MORE INPUTTED CDR LOCATION PATH MODIFICATIONS
712
UPDATE ONE OR MORE CONFIGURATION PARAMETERS IN THE BUSINESS LOGIC BASED ON ONE OR MORE INPUTTED PARAMETER UPDATES
714
RECEIVE A SELECTION OF ONE OR MORE CDR DISTRIBUTOR APPLICATIONS OF A PLURALITY OF CDR DISTRIBUTOR APPLICATIONS LISTED IN THE BUSINESS LOGIC
A
724
TERMINATE RECALCULATION OF THE SERVICE FEES CHARGED TO THE PARTICULAR SET OF SUBSCRIBERS

FIG. 7a

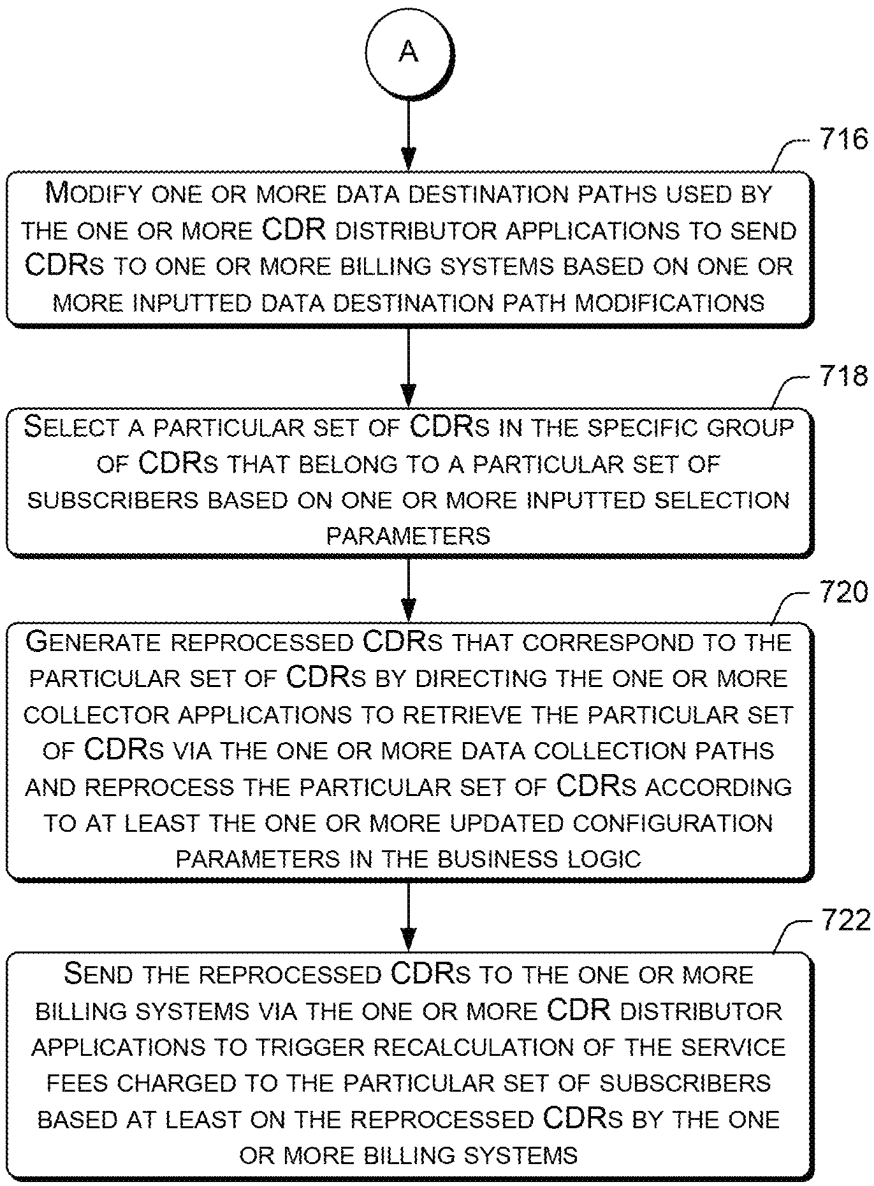

A
716
MODIFY ONE OR MORE DATA DESTINATION PATHS USED BY THE ONE OR MORE CDR DISTRIBUTOR APPLICATIONS TO SEND CDRS TO ONE OR MORE BILLING SYSTEMS BASED ON ONE OR MORE INPUTTED DATA DESTINATION PATH MODIFICATIONS
718
SELECT A PARTICULAR SET OF CDRS IN THE SPECIFIC GROUP OF CDRS THAT BELONG TO A PARTICULAR SET OF SUBSCRIBERS BASED ON ONE OR MORE INPUTTED SELECTION PARAMETERS
720
GENERATE REPROCESSED CDRS THAT CORRESPOND TO THE PARTICULAR SET OF CDRS BY DIRECTING THE ONE OR MORE COLLECTOR APPLICATIONS TO RETRIEVE THE PARTICULAR SET OF CDRS VIA THE ONE OR MORE DATA COLLECTION PATHS AND REPROCESS THE PARTICULAR SET OF CDRS ACCORDING TO AT LEAST THE ONE OR MORE UPDATED CONFIGURATION PARAMETERS IN THE BUSINESS LOGIC
722
SEND THE REPROCESSED CDRS TO THE ONE OR MORE BILLING SYSTEMS VIA THE ONE OR MORE CDR DISTRIBUTOR APPLICATIONS TO TRIGGER RECALCULATION OF THE SERVICE FEES CHARGED TO THE PARTICULAR SET OF SUBSCRIBERS BASED AT LEAST ON THE REPROCESSED CDRS BY THE ONE OR MORE BILLING SYSTEMS

FIG. 7b

FEE RE-RATING FOR A WIRELESS CARRIER NETWORK

BACKGROUND

The subscribers of a wireless carrier network are billed service fees for using the wireless telecommunication services provided by the wireless carrier network. For example, the wireless telecommunication services may include voice calling, text or multimedia messaging, and uploading and downloading of data via the Internet using various wireless telecommunication devices. Depending on the wireless service plan of a subscriber, the service fees charged to the subscriber may include recurring service fees and/or one-time service fees. Such service fees may be calculated by the wireless carrier network based on service usage records, also referred to as call detail records (CDRs), that are generated by the various network nodes of the wireless carrier network. For example, such network nodes of the wireless carrier network may include Charge and Control nodes (CCNs), Operations Control Centers (OCCs), IP Multimedia Subsystems (IMSs), Affinity Network Fusion (ANF) components, Similarity Network Fusion (SNF) components, and/or so forth. The CDRs are then used by the various billing systems of the wireless carrier network to generate service fee bills for the subscribers. The term "re-rating" refers to a process by which such service fees of a subscriber may be recalculated so that the service fees may be adjusted after the service fees are initially calculated by a billing system.

BRIEF DESCRIPTION OF THE DRAWINGS

The detailed description is described with reference to the accompanying figures, in which the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items.

FIG. **1** illustrates an example architecture that enables the use of a re-rating engine to initiate the re-rating of service fees for subscribers by the billing systems of a wireless carrier network.

FIG. **2** is a block diagram showing various components of one or more computing devices that support the implementation of a re-rating engine that is capable of initiating the re-rating of service fees for subscribers by the billing systems of a wireless carrier network.

FIG. **3***a* illustrates an example database query interface provided by a lookup tool of the re-rating engine for querying a database for lists of subscriber identifiers that are affected by an anomalous event.

FIG. **3***b* illustrates an example database query result interface provided by the lookup tool of the re-rating engine that presents a list of subscriber identifiers for exporting to a CDR reprocessing tool such that reprocessed CDRs can be generated for the list of sub scriber identifiers.

FIG. **4***a* illustrates an example record query interface provided by the lookup tool of the re-rating engine for querying a database for a list of CDRs that are affected by an anomalous event.

FIG. **4***b* illustrates an example record query result interface provided by the lookup tool of the re-rating engine that presents the list of CDRs for exporting to the CDR reprocessing tool such that reprocessed CDRs can be generated for the list CDRs.

FIG. **5***a* illustrates an example automated query interface provided by the lookup tool of the re-rating engine for querying a database for a list of subscriber identifiers or a list of CDRs that are affected by an anomalous event.

FIG. **5***b* illustrates an example automated query result interface provided by the lookup tool of the re-rating engine that presents the list of subscriber identifiers or the list of CDRs for exporting to the CDR reprocessing tool such that reprocessed CDRs can be generated.

FIG. **6** is a flow diagram of an example process for monitoring one or more network nodes of a wireless carrier network to detect an anomalous event.

FIGS. **7***a* and **7***b* illustrate a flow diagram of an example process for using a re-rating engine to initiate the re-rating of service fees for subscribers by the billing systems of a wireless carrier network.

DETAILED DESCRIPTION

This disclosure is directed to techniques that enable a wireless carrier network to re-rate the service fees that are owed by the subscribers for using the wireless telecommunication such that the service fees may be adjusted after the service fees are initially calculated. The service fees may be initially calculated based on service usage records, also referred to as call detail records (CDRs), that are generated by the various network nodes of the wireless carrier network.

However, in some instances, the services fees may have to be recalculated for a set of subscribers. In order for the service fees to be recalculated for the set of subscribers, the relevant CDRs that are associated with the set of subscribers may be identified. For example, a re-rating engine may include a lookup tool that enables an administrator to query for affected CDRs of subscribers that fall within a particular date range and fit certain search attributes (e.g., account type, service class, alarm identifier, etc.) via a database lookup, a record lookup, or an automated lookup. Subsequently, a business logic that was originally used to initiate the generation of the service fees for the set of subscribers by one or more billing systems based on the affected CDRs may be modified. Such modification may be performed using a CDR reprocessing tool of the re-rating engine. Once the business logic is modified, the CDR reprocessing tool may use the modified business logic to initiate the generation of reprocessed CDRs that correspond to the affected CDRs. The reprocessed CDRs are then sent to the one or more billing systems according to the business logic to trigger the recalculation of the service fees for the set of subscribers.

In various embodiments, the services fees of the subscribers may have to be recalculated for a set of subscribers because of an initial error in the business logic, because of configuration errors or failures of the billing systems, because the wireless carrier network wants to provide discounted service or free service to subscribers affected by a disaster, or because of some other anomalous event occurred.

Thus, the re-rating engine may provide an administrator of the wireless carrier network with an efficient way to identify and modify large batches of CDRs (e.g., hundreds of thousands CDRs) so that service fees can be expediently recalculated following an anomalous event. The techniques described herein may be implemented in a number of ways. Example implementations are provided below with reference to the following figures.

Example Architecture

FIG. **1** illustrates an example architecture **100** that enables the use of a re-rating engine to initiate the re-rating of service fees for subscribers by the billing systems of a wireless carrier network. The architecture **100** illustrates selected network components of a wireless carrier network. In various embodiments, the wireless carrier network may include a core network and a radio access network. The radio access network may include multiple base stations. Each base station may include a base transceiver system (BTS) that communicates via an antenna system over an air-link with one or more user devices that are within range. The BTS may send radio communication signals to user devices and receive radio communication signals from user devices. The base stations may provide corresponding network cells that deliver telecommunication and data communication coverage to user devices. The user devices may include a smartphone, a tablet computer, an embedded computer system, or any other device that is capable of using the wireless communication services that are provided by the wireless carrier network.

The core network may use the network cells to provide wireless communication services to the user devices. The core network may include components that support 3G voice communication traffic, as well as 3G, 4G, and 5G data communication traffic. For example, 3G data communication traffic between a user device and the Internet may be routed through a gateway of a 3G Packet Switch (PS) Core. On the other hand, 3G voice communication traffic between the user device and a Public Switched Telephone Network (PSTN) may be routed through a Mobile Switch (MSC) of a 3G Circuit Switch (CS) core. The core network may further include components that support 4G and 5G voice and data communication traffic. Such components may include an Evolved Packet Core (EPC) and an IP Multimedia Subsystem (IMS) core. The IMS core may provide the user devices with data access to external packet data networks, such as the networks of other wireless telecommunication providers, as well as backend servers in the core network. The EPC may include a Mobility Management Entity (MME). The MME may handle paging, authentication, and registration of user devices, as well as the routing of data and voice communications through selected gateways.

The wireless carrier network may further include mediation servers **102** that generate call detail records (CDRs) for various network components that provide wireless telecommunication services to subscribers. For example, the mediation servers **102** may be parts of Charge and Control nodes (CCNs), Operations Control Centers (OCCs), IP Multimedia Subsystems (IMSs), Affinity Network Fusion (ANF) components, Similarity Network Fusion (SNF) components, and/or so forth. For example, a first mediation server may generate CDRs related to the use of 5G voice calls by subscribers, a second mediation server may generate CDRs related to the upload/download data via the wireless carrier network by subscribers, a third mediation server may generate CDRs related to the sending/receiving of text or multimedia message by subscribers, and/or so forth. In some instances, mediation servers may be responsible for generating CDRs for not only subscribers of the wireless carrier network, but also subscribers of mobile virtual network operators (MVNO) that provide telecommunication services to their subscribers via the wireless carrier network.

In operation, a central mediation engine **104** that is executed by one or more computing devices **106** of the wireless carrier network may use a collection layer **108** to collect the CDRs from the mediation servers **102**. One example of the central mediation engine **104** is the Ericsson Multi-Mediation (EMM) engine developed by Telefonaktiebolaget LM Ericsson (Ericsson) of Stockholm, Sweden. The collection layer **108** may include corresponding collector applications that interact with the mediation servers to retrieve the CDRs from the mediation servers **102**. For example, a collector application may be configured to navigate to a CDR data store of a mediation server via a corresponding collection path (e.g., a file directory path) and retrieve the CDRs at designated dates and times according to a set of configuration parameters. Such configuration parameters for the retrieval of the CDRs by the collector application may be stored in a business logic that is implemented by the central mediation engine **104**. The retrieval of the CDRs may be accomplished using file transfer protocol (FTP), SSH file transfer protocol (SFTP), Hypertext Transfer Protocol (HTTP) posts, direct network infrastructure element connection, and/or other data transfer techniques.

In some embodiments, the CDRs may be processed by the collector applications following retrieval from the mediation servers. For example, the processing of the CDRs may include reformatting the CDRs, filtering out specific data from the CDRs, or adding additional data to the CDRs. Accordingly, the collector applications may process the CDRs according to configuration parameters specified by a business logic that is implemented by the central mediation engine **104**. The processed CDRs are then sent by the central mediation engine **104** via a distribution layer **110** to one or more billing systems, such as the billing systems **112**(**1**)-**112**(N).

The distribution layer **110** may include corresponding distribution applications that deliver retrieved and/or processed CDRs to one or more of the billing systems **112**(**1**)-**112**(N). In turn, the one or more billing systems **112**(**1**)-**112**(N) may generate bills of service fees to be charged to subscribers. For example, a distribution application may be configured to send a set of corresponding CDR from a specific mediation server to a data store of a particular billing system at particular dates and times via a destination path (e.g., a file directory path) according to one or more configuration parameters. Such configuration parameters related to the distribution of the CDRs by the collector application may be stored in the business logic that is implemented by the central mediation engine **104**. In various embodiments, a business logic may be in the form of a configuration file that stores various configuration parameters. The configuration parameters may be encoded by any data format, such as Extensible Markup Language (XML). In this way, a business logic may store configuration parameters related to retrieving CDRs from one or more specific mediation servers, processing the CDRs into specific data formats, and delivering the CDRs to one or more specific billing systems.

In some scenarios, an administrator **114** of the wireless carrier network may need to modify the service fees that are charged to particular subscribers. For example, the services fees may have to be recalculated for a set of subscribers because of an initial error in the business logic, because of configuration errors or failures of the billing systems, because the wireless carrier network wants to provide discounted service or free service to subscribers affected by a disaster, or because of some other anomalous event occurred. In some instances, the anomalous event may result in a customer billing complaint (e.g., text or call to customer service) that is ultimately routed to the administrator. In other instances, the anomalous event may result in an automated system alarm that notifies the administrator of the event. In such scenarios, the administrator **114** may use a re-rating engine **116** that is executed by the one or more computing devices **106** to perform the modification of the service fees. The administrator **114** may use a client application **118** on a client device **120** to interact with the re-rating engine **116**. The re-rating engine **116** may provide application user interfaces that are accessible via the client application **118**. The application user interfaces may be web pages or interface pages that are accessible via the client application **118**. For example, the web pages may be in the form of HyperText Markup Language (HTML) documents, in which the documents may include text content, images, multimedia content, cascade style sheets (CSS), and/or scripts that are accessible via a web browser. In some embodiments, the re-rating engine **116** may use an application server that supports server-side scripting via multiple scripting languages, such as Active Server Pages (ASP), Hypertext Preprocessor (PHP), JavaScript, and other scripting languages to support the dynamic generation of web pages that presents output and receives input from a user.

Accordingly, the administrator **114** may use a lookup tool **122** of the re-rating engine **116** to query for CDRs that match one or more query criteria. A CDR may contain fields that are organized in a structured manner to provide information regarding a communication session that occurred. For example, A CDR for a voice call may include a first field that contains a MSISDN of a caller, a second field that contains a MSISDN of a callee, a third field that contains a date and time that the voice call is made, and a fourth field that contains a call duration, and/or so forth. The various fields in the CDR may be parsed and viewed via a formatter function in the business logic. The one or more query criteria may correspond to one or more characteristics of an anomalous event that occurred. For example, the query criteria may include a particular date range and one or more search attributes, such as account type, service class, alarm identifier, and/or so forth. In some instances, the lookup tool **122** may be used to perform a subscriber lookup **124** to retrieve subscriber identifiers (e.g., MSISDNs) of subscribers that are impacted by an anomalous event from system data stores **126**. The system data stores **126** may include data stores that are maintained by the billing, tariff, and/or rating systems of the wireless carrier network. In other instances, the lookup tool **122** may be used to perform a CDR lookup **128** to retrieve CDR identifiers (e.g., CDR file names) of CDRs that correspond to an anomalous event from a call data recordkeeping data store **130**. For example, a CDR file identified by a CDR identifier may contain the CDRs that are collected by a collector on a particular day. The call data recordkeeping data store **130** may be a data archive store that is configured to archive CDRs of the wireless carrier network. The call data recordkeeping data store **130** may be configured to store each CDR for a predetermined period of time (e.g., three months) before the CDR is purged from the data archive store by a data maintenance function of the call data recordkeeping data store **130**.

In additional instances, the lookup tool **122** may be used to perform an application program interface (API) request lookup **132** to retrieve subscriber identifiers of subscribers and/or CDR identifiers of CDRs that are related to an alarm event from the data stores of the mediation servers **102**. For example, the alarm event may be an anomalous event that is automatically detected by an event detection engine **134** executing on the one or more computing devices **106**. The event detection engine **134** may be an application that monitors various rating, billing, and tariff components of the wireless carrier networks for failures related to the generation of CDRs and bills for subscribers. Such failures may include software failures (e.g., such as failures due to the execution of faulty software code or faulty configuration settings) and/or hardware failures, such as malfunction of servers that execute software. Accordingly, the API request lookup **132** may be used to retrieve subscriber identifiers and/or CDR identifiers that correspond to different alarm events.

Following the retrieval of the subscriber identifiers and/or CDR identifiers that are related to an anomalous event via the lookup tool **122**, the administrator **114** may use the lookup tool **122** to export the subscriber identifiers and/or CDR identifiers as input parameters **136** to a CDR reprocessing tool **138** of the re-rating engine **116** to initiate a reprocessing of the corresponding CDRs. The reprocessed CDRs, such as the reprocessed CDRs **140(1)**-**140**(N), are then sent to the respective billing systems, such as the billing systems **112(1)**-**112**(N), may perform re-rating to recalculate service fees for the subscriber that are affected by the anomalous event.

In various embodiments, the CDR reprocess tool **138** may provide multiple application user interfaces that enable the administrator **114** to control the reprocessing of a set of CDRs. The application user interfaces may provide menus, icons, selection boxes, selection buttons, and/or other interface elements to receive input from the administrator **114** and output information to the administrator **114**. The set of CDRs may include CDRs that are identified by the CDR identifiers or related to the subscriber identifiers that are exported by the lookup tool **122**. Accordingly, the CDR reprocess tool **138** may be used by the administrator **114** to import a business logic that is used to generate the set of CDRs. The business logic may be visually presented by the CDR reprocess tool **138** to the administrator **114** via a logic tree diagram, in which the relationship between functions (e.g., filter function, format function, etc.), sub-functions of the functions, configuration parameters for the functions and/or sub-functions, and related sub-configuration parameters in the business logic are presented as interconnected and/or hierarchical branches and nodes. Alternatively, the business logic may be presented by the CDR reprocess tool **138** in text form as a script that contains configuration codes that correspond to the functions, sub-functions, configuration parameters, and sub-configuration parameters.

Subsequently, the CDR reprocess tool **138** may receive a selection of one or more data collector applications of a plurality of data collector applications that are listed in the business logic. The selection may be inputted by the administrator **114** via an application user interface. In various instances, the administrator **114** may select the one or more data collector applications based on the knowledge the administrator **114** possesses regarding the network architecture of the wireless carrier network. For example, the administrator **114** may select collector applications that are associated with specific CCNs, OCCs, etc. In another example, the administrator **114** may select collector applications of network components that service particular categories of subscribers, such as the subscribers of an MVNO that is affiliated with the wireless carrier network.

Following the selection, the CDR reprocess tool **138** may determine whether a specific group of CDRs is currently archived in a data archive store of the wireless carrier network, such as the call data recordkeeping data store **130**. For example, the specific group of CDRs may be CDRs that have been identified by the administrator **114** using the lookup tool **122** as to be re-rated. These CDRs may be identified to the CDR reprocess tool **138** via CDR identifiers imported from the lookup tool **122**. Alternatively, These CDRs may be identified to the CDR reprocess tool **138** via corresponding subscriber identifiers imported from the lookup tool **122**. In such alternative instances, the CDR reprocess tool **138** may use an identification function to identify CDRs that contain service usage details for the subscribers identified by the imported subscriber identifiers. The specific group of CDRs must be archived in the data archive store in order for the CDR reprocess tool **138** to perform the reprocessing. Thus, if the specific group of CDRs is present in the data archive store, the administrator **114** may use the CDR reprocess tool **138** to optionally modify one or more CDR location paths to the specific group of CDRs in the data archive store by inputting one or more one or more inputted CDR location path modifications. A CDR location path may be a network file directory path to a file location where at least one particular CDR is stored. For example, the administrator **114** may recognize from prior knowledge that a CDR location path is incorrect and input the corrective CDR location path modification. The one or more modified CDR location paths are then stored in the business logic by the CDR reprocessing tool.

Additionally, the CDR reprocess tool **138** may receive one or more parameter updates for the one or more configuration parameters in the business logic, in which the one or more parameter updates are inputted by the administrator **114**. The one or more parameter updates may cause the CDR reprocess tool **138** to modify the configuration parameters in the business logic so that the resultant reprocessed CDRs will eventually cause a billing system to modify the service fees charged to subscribers as compared to the previously processed CDRs. For example, a parameter update may change a type of service designation for a telecommunication service usage session detailed in the CDRs, and another parameter update may adjust a date and/or time of a telecommunication service usage session detailed in the CDRs. These and other parameter updates may be used to adjust a service fee (e.g., increase, reduced, or zeroed-out) for a telecommunication service usage session of a subscriber, or adjust a service fee charged to the subscriber for a particular type of wireless telecommunication service.

Along with the modifications of the configuration parameters in the business logic, the CDR reprocess tool **138** may further receive a selection of one or more CDR distribution applications of a plurality of CDR distributor applications listed in the business logic. The selection may be inputted by the administrator **114** via an application user interface. In various instances, the administrator **114** may select the one or more CDR distributor applications based on the knowledge the administrator **114** possesses regarding the network architecture of the wireless carrier network. For example, the administrator **114** may select CDR distributor applications that distribute CDRs to specific billing systems, such as billing systems that generate service fees for specific network components or categories of subscribers. In some instances, the data destination paths used by the one or more selected CDR distributor applications to send CDRs to the corresponding billing systems, which are stored in the business logic, may be optionally modified. A data destination path may be a network file directory path to a file location where the CDRs are subsequently retrieved and accessed by the billing system. For example, the administrator **114** may recognize from prior knowledge that a data destination path is incorrect and input a corrective CDR location path modification.

In some instances, the CDR reprocess tool **138** may further filter out one or more of the CDRs from the specific group of CDRs before reprocessing. In such instances, the CDR reprocess tool **138** may select a particular set of CDRs in the specific group of CDR based on one or more inputted selection parameters, in which the particular set of CDRs may correspond to a particular set of subscribers. The selection parameters may be inputted by the administrator **114** via an application user interface of the CDR reprocess tool **138**. The inputted selection parameters may be used by a filter function of the business logic to filter out CDRs that do not meet the inputted selection parameters. For example, the selection parameters may include a CDR date and time range, a range of subscriber identifiers (MSISDNs), a range of subscriber identifiers (MSISDNs) with certain characteristics (e.g., active, inactive), identifiers of CDRs and/or subscribers that correspond to one or more specific alarm events, a list of specific subscriber identifiers that is imported into the tool, a list of specific CDR identifiers that is imported into the tool, and/or so forth. These selections and modifications may result in an updated business logic that is used for generating reprocessed CDRs.

Subsequently, the CDR reprocess tool **138** may be activated via an application user interface to generate the reprocessed CDRs using the updated business logic. For example, the administrator **114** may use an activation button on the application user interface to initiate the reprocessing. Following activation, the CDR reprocess tool **138** may direct the one or more selected collector applications to retrieve and access the particular set of CDRs via the one or more data collection paths such that the collector applications may reprocess the CDRs according to at least the one or more updated configuration parameters in the business logic. In various embodiments, the selected collector applications may use the updated configuration parameters in the updated business logic to format and generate the reprocessed CDRs. The reprocessed CDRs are then sent by the one or more selected CDR distributor applications as directed by the updated business logic to the one or more billing systems. In turn, the reprocessed CDRs may trigger recalculation of the service fees charged to the particular set of subscribers by the one or more billing systems based at least on the reprocessed CDRs.

Example Computing Device Components

FIG. **2** is a block diagram showing various components of one or more computing devices that support the implementation of a re-rating engine that is capable of initiating the re-rating of service fees for subscribers by the billing systems of a wireless carrier network. The computing devices **106** may include a communication interface **202**, one or more processors **204**, memory **206**, and hardware **208**. The communication interface **202** may include wireless and/or wired communication components that enable the computing devices **106** to transmit data to and receive data from other networked devices. The hardware **208** may include additional user interface, data communication, or data storage hardware. For example, the user interfaces may include a data output device (e.g., visual display, audio speakers), and one or more data input devices. The data input devices may include, but are not limited to, combinations of one or more of keypads, keyboards, mouse devices, touch screens that accept gestures, microphones, voice or speech recognition devices, and any other suitable devices.

The memory **206** may be implemented using computer-readable media, such as computer storage media. Computer-readable media includes, at least, two types of computer-readable media, namely computer storage media and communications media. Computer storage media includes volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD), high-definition multimedia/data storage disks, or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other non-transmission medium that can be used to store information for access by a computing device. In contrast, communication media may embody computer-readable instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave, or other transmission mechanisms. In other embodiments, the computing devices **106** or components thereof may be virtual computing devices in the form of virtual machines or software containers that are hosted in a computing cloud.

The computing devices **106** may implement various back-end services and components of the wireless carrier network that are associated with performing re-rating, such as the re-rating engine **116**, the event detection engine **134**, and the central mediation engine **104**. The implementation involves the execution of software, applications, and/or modules that include routines, program instructions, code segments, objects, and/or data structures that perform particular tasks or implement particular abstract data types. Further, the various data stores of the wireless carrier network may include databases that are implemented in the computing cloud or on-premise in one or more facilities of the wireless carrier network.

Example Application User Interfaces

FIG. **3***a* illustrates an example database query interface provided by a lookup tool of the re-rating engine for querying a database for lists of subscriber identifiers that are affected by an anomalous event. The database query interface **300** may enable a subscriber lookup of subscriber identifiers (e.g., MSISDNs) from a data store of the wireless carrier network based on one or more inputted query parameters. For example, the query parameters that can be selected from multiple available options may include a specific node identifier **302**, a specific telecommunication service parameter value **304**, and/or a specific date range **306**. Following the input of various query parameters, the user may activate a query button **308** to direct the lookup tool **122** to execute the query. The database query interface **300** may further include a query writeup area **310** that enables a user to compose a custom database query for specific subscriber identifiers. Following the input of a custom query, the user may activate a query button **312** to direct the lookup tool **122** to execute the query. Accordingly, the database query interface **300** may enable the administrator **114** to look up subscriber identifiers of subscribers that are affected by an anomalous event via one or more inputted query parameters that the administrator **114** knows are associated with the event.

FIG. **3***b* illustrates an example database query result interface provided by the lookup tool of the re-rating engine. The database query result interface **314** may be presented by the lookup tool **122** following the execution of a database query. The database query result interface **314** may show one or more lists of subscriber identifiers (e.g., MSISDNs) of subscribers that match the query parameters of the database query. For example, the lists may include an overall list of subscribers **316**, a list of inactive subscribers **318**, and a list of active subscribers **320**. The windows that present the lists **316**, **318**, and **320** may be equipped with corresponding export buttons **322**, **324**, and **326**. Thus, the activation of an export button may cause the lookup tool **122** to export a corresponding list of subscriber identifiers to the CDR reprocessing tool **138** such that reprocessed CDRs may be generated.

FIG. **4***a* illustrates an example record query interface provided by the lookup tool of the re-rating engine for querying a database for a list of CDRs that are affected by an anomalous event. The record query interface **400** may enable a record lookup of CDR identifiers (e.g., CDR file names) from a data store of the wireless carrier network based on one or more inputted query parameters. For example, the query parameters that can be selected from multiple available options may include a specific server/node identifier **402**, a specific CDR name **404**, a specific directory path **406** where the CDR are located, and a specific date range **408**. Following the input of various query parameters, the user may activate a query button **410** to direct the lookup tool **122** to execute the query. Accordingly, the record query interface **400** may enable the administrator **114** to look up CDR identifiers of CDRs that are affected by an anomalous event via one or more inputted query parameters that the administrator **114** knows are associated with the event.

FIG. **4***b* illustrates an example record query result interface provided by the lookup tool of the re-rating engine. The record query result interface **412** may be presented by the lookup tool **122** following the execution of a record query. The record query result interface **412** may show one or more lists of CDR identifiers (e.g., file names) that match the query parameters of the database query in a query result window **414**. The query result window **414** may be configured to enable a user to select one or more CDR identifiers. Following the selection of the one or more CDR identifiers, the user may activate an export button **416** to export the one or more selected CDR identifiers to the CDR reprocessing tool **138**, such that reprocessed CDRs may be generated for at least some of the CDRs identified by the one or more selected CDR identifiers.

FIG. **5***a* illustrates an example automated query interface provided by the lookup tool of the re-rating engine for querying a database for a list of subscriber identifiers or a list of CDRs that are affected by an anomalous event. The automated query interface **500** may enable a lookup of subscriber identifiers (e.g., MSISDNs) and/or CDR identifiers (e.g., CDR file names) that are associated with an alarm event from a data store of the wireless carrier network based on one or more inputted query parameters. For example, the query parameters that can be selected from multiple available options may include a specific server identifier **502**, an alarm identifier/code identifier **504**, and a specific date range **506**. The selection of a specific alarm identifier/code identifier may cause an error instance window **508** to show multiple error instances that are associated with an alarm. The error instance window **508** may further enable a user to select a specific error instance file so that additional detailed information regarding the error instance may be presented in the detail information window **510**. Further, following the selection of the error instance file, the user may activate a query button **512** to direct the lookup tool **122** to execute a query on the error instance.

FIG. **5***b* illustrates an example automated query result interface provided by the lookup tool of the re-rating engine. The automated query result interface **514** may be presented by the lookup tool **122** following the execution of a query via the automated query interface. The query result interface **514** may show one or more lists of subscriber identifiers (e.g., MSISDNs) of subscribers that match the query parameters of the database query. For example, the lists may include an overall list of subscribers **516**, a list of inactive subscribers **518**, and a list of active subscribers **520**. The windows that present the lists **516**, **518**, and **520** may be equipped with corresponding export buttons **522**, **524**, and **526**. Thus, the activation of an export button may cause the lookup tool **122** to export a corresponding list of subscriber identifiers to the CDR reprocessing tool **138** such that reprocessed CDRs may be generated for the corresponding list.

Additionally, the automated query result interface **514** may show one or more lists of CDR identifiers (e.g., file names) that match the query parameters of the database query in a CDR list window **528**. The CDR list window **528** may be configured to enable a user to select one or more CDR identifiers. Following the selection of the one or more CDR identifiers, the user may activate an export button **530** to export the one or more selected CDR identifiers to the CDR reprocessing tool **138**, such that reprocessed CDRs may be generated for at least some of the CDRs identified by the one or more selected CDR identifiers.

Example Processes

FIGS. **6**, **7***a*, and **7***b* present illustrative processes **600** and **700** for initiating the re-rating of service fees for subscribers by the billing systems of a wireless carrier network. Each of the processes **600** and **700** is illustrated as a collection of blocks in a logical flow chart, which represents a sequence of operations that can be implemented in hardware, software, or a combination thereof. In the context of software, the blocks represent computer-executable instructions that, when executed by one or more processors, perform the recited operations. Generally, computer-executable instructions may include routines, code segments, programs, objects, components, data structures, and the like that perform particular functions or implement particular abstract data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks can be combined in any order and/or in parallel to implement the process. For discussion purposes, the processes **600** and **700** are described with reference to the architecture **100** of FIG. **1**.

FIG. **6** is a flow diagram of an example process **600** for monitoring one or more network nodes of a wireless carrier network to detect an anomalous event. At block **602**, an event detection engine **134** may monitor one or more network components of a wireless carrier network. The event detection engine **134** may be an application that monitors various rating components of the wireless carrier networks for failures related to the generation of CDRs and bills for subscribers. Such failures may include software failures (e.g., such as failures due to the execution of faulty software code or faulty configuration settings) and/or hardware failures, such as malfunction of servers that execute software.

At block **604**, the event detection engine **134** may detect that an anomalous event occurred with respect to at least one network component of the wireless carrier network. For example, the detection of the anomalous event may cause the event detection engine **134** to generate an alarm event that is identifiable via one or more error instances. At block **606**, the re-rating engine **116** may identify a particular set of CDRs associated with the group of subscribers that are affected by the anomalous event. In various embodiments, the particular set of CDRs may be identified via CDR identifiers that are associated with one or more error instances, or corresponding subscriber identifiers that are associated with the one or more error instances. The identification of the particular set of CDRs may be performed automatically by the re-rating engine **116** or based on query parameters inputted via the client application **118**.

At block **608**, the re-rating engine **116** may modify the service fees charged to the group of subscribers by generating reprocessed CDRs that correspond to the particular set of CDRs for processing by one or more billing systems. In various embodiments, the block **608** may be implemented via the re-rating engine **116** for a group of subscribers for reasons other than the detection of an anomalous event internal to the wireless carrier network. For example, the modification may be implemented due to an occurrence of a natural disaster that negatively impacts the group of subscribers, and the wireless carrier may decide to provide discounted or free telecommunication services for a specific period of time to the group of subscribers.

FIGS. **7***a* and **7***b* illustrate a flow diagram of an example process **700** for using a re-rating engine to initiate the re-rating of service fees for subscribers by the billing systems of a wireless carrier network. The process **700** may further illustrate block **608** of the process **600**. At block **702**, the re-rating engine **116** may import a business logic that is used to generate CDRs for a plurality of subscribers that use wireless telecommunication services provided by a wireless carrier network. In various embodiments, the business logic may be in the form of a configuration file that stores various configuration parameters. At block **704**, the re-rating engine **116** may receive a selection of one or more data collector applications of a plurality of data collector applications listed in the business logic. In various embodiments, the data collector applications may be included in a collector layer and are configured to retrieve CDRs from data stores of various servers, such as mediation servers, and process them according to the configuration parameters in the business logic.

At block **706**, the re-rating engine **116** may determine whether a specific group of CDRs are currently archived in a data archive store of the wireless carrier network. The data archive store may be configured to store a set of CDRs for a predetermined period of time (e.g., three months) before the set of CDRs is purged from the data archive store by a data maintenance function of the data archive store.

At decision block **708**, if the specific group of CDRs is currently archived in the data archive store of the wireless carrier network ("yes" at decision block **708**), the process **700** may proceed to block **710**. At block **710**, the re-rating engine **116** may modify one or more CDR location paths to the specific group of CDRs in the data archive store based on one or more inputted CDR location path modifications. The block **710** may be optional and the one or more CDR location paths are not modified by the re-rating engine **116** when the re-rating engine **116** does not receive any CDR location path modification inputs.

At block **712**, the re-rating engine **116** may update one or more configuration parameters in the business logic based on one or more inputted parameter updates. For example, a parameter update may change a type of service designation for the telecommunication service usage session detailed in the CDRs, another parameter update may adjust a date and/or time of the telecommunication service usage session detailed in the CDRs, and/or so forth.

At block **714**, the re-rating engine **116** may receive a selection of one or more CDR distributor applications of a plurality of CDR distributor applications listed in the business logic. The one or more CDR distribution applications may be included in a distributor layer and are configured to deliver the retrieved and/or processed CDRs to one or more of the billing systems. At block **716**, the re-rating engine **116** may modify one or more data destination paths used by the one or more CDR distributor applications to send CDRs to one or more billing systems based on one or more inputted data destination path modifications. The block **716** may be optional and the one or data destination paths are not modified by the re-rating engine **116** when the re-rating engine **116** does not receive any data destination path modification inputs.

At block **718**, the re-rating engine **116** may select a particular set of CDRs in the specific group of CDRs that belong to a particular set of subscribers based on one or more inputted selection parameters. For example, the selection parameters may include a CDR date and time range, a range of subscriber identifiers (MSISDNs) with certain characteristics (e.g., active, inactive, etc.), CDRs or subscribers that correspond to one or more specific alarm events, a list of specific subscriber identifiers that is imported into the tool, a list of specific CDR identifiers that is imported into the tool, and/or so forth.

At block **720**, the re-rating engine **116** may generate reprocessed CDRs that correspond to the particular set of CDRs by directing one or more collector applications to retrieve the particular set of CDRs from the data archive store via the one or more data collection paths and reprocess the particular set of CDRs according to at least the one or more updated configuration parameters in the business logic.

At block **722**, the re-rating engine **116** may send the reprocessed CDRs to the one or more billing systems via the one or more CDR distributor applications to trigger recalculation of service fees charged to the particular set of subscribers based at least on the reprocessed CDRs by the one or more billing systems. Returning to decision block **708**, if the specific group of CDRs is not archived in the data archive store of the wireless carrier network ("no" at decision block **708**), the process **700** may proceed to block **724**. At block **724**, the re-rating engine **116** may terminate the recalculation of the service fees charged to the particular set of subscribers.

CONCLUSION

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as exemplary forms of implementing the claims.

What is claimed is:

1. One or more non-transitory computer-readable media storing computer-executable instructions that upon execution cause one or more processors to perform acts that comprise re-rating acts, the re-rating acts including:

importing a business logic that is used to generate call details records (CDRs) for a plurality of subscribers that use wireless telecommunication services provided by a wireless carrier network;

receiving a selection of one or more data collector applications of a plurality of data collector applications that are listed in the business logic, the one or more data collector applications corresponding to a specific group of CDRs, the plurality of data collector applications configured to collect CDRs from one or more servers of the wireless carrier network as the plurality of subscribers use the wireless telecommunication services;

updating one or more configuration parameters in the business logic based on one or more inputted parameter updates;

receiving a selection of one or more CDR distributor applications of a plurality of CDR distributor applications listed in the business logic, the plurality of CDR distributor applications configured to distribute the CDRs to one or more billing systems of the wireless carrier network;

determining that an anomalous event that is external to the wireless carrier network occurred and negatively impacted a particular set of subscribers;

based on the anomalous event that is external to the wireless carrier network occurring and negatively impacting the particular set of subscribers, determining to modify the service fees charged to the particular set of subscribers;

selecting a particular set of CDRs in the specific group of CDRs that belong to the particular set of subscribers based on one or more inputted selection parameters;

generating reprocessed CDRs that correspond to the particular set of CDRs by directing the one or more collector applications to retrieve the particular set of CDRs from a data archive store of the wireless carrier network and reprocess the particular set of CDRs according to at least the one or more updated configuration parameters in the business logic; and

sending the reprocessed CDRs to the one or more billing systems via the one or more CDR distributor applications to trigger recalculation of service fees charged to the particular set of subscribers based at least on the reprocessed CDRs by the one or more billing systems.

2. The one or more non-transitory computer-readable media of claim **1**, further comprising acts that include:

querying one or more data stores of the wireless carrier network to identify the particular set of subscribers; and

performing the re-rating acts to modify the service fees charged to the particular set of subscribers.

3. The one or more non-transitory computer-readable media of claim **2**, wherein the querying includes retrieving subscriber identifiers of the particular set of subscribers via a subscriber lookup query, CDR identifiers of CDRs corresponding to the particular set of subscribers via a record lookup query, or at least one of the subscriber identifiers or CDR identifier via an automated alarm event query.

4. The one or more non-transitory computer-readable media of claim **3**, wherein the subscriber identifiers of the subscribers include mobile station international subscriber directory number (MSISDNs).

5. The one or more non-transitory computer-readable media of claim **2**, wherein the querying includes querying in response to detecting an occurrence of the anomalous event.

6. The one or more non-transitory computer-readable media of claim **5**, wherein the anomalous event is an occurrence of a natural disaster in a geographical area that affects the particular set of subscribers.

7. The one or more non-transitory computer-readable media of claim **1**, wherein the one or more inputted parameter updates include at least one parameter update that adjusts a service fee charged to a subscriber or adjust a service fee charged to the subscriber for a particular type of wireless telecommunication service.

8. The one or more non-transitory computer-readable media of claim **1**, wherein the business logic is encoded via Extensible Markup Language (XML) in a corresponding configuration file.

9. The one or more non-transitory computer-readable media of claim 1, wherein the re-rating acts further include determining whether the specific group of CDRs is currently archived in the data archive store of the wireless carrier network in addition to being previously sent to the one or more billing systems, and wherein the updating the one or more configuration parameters, the receiving a selection of one or more CDR distributor applications, the selecting the particular set of CDRs, and the generating the reprocessed CDRs, and the sending the reprocessed CDRs are performed when the specific group of CDRs are archived in the data archive store.

10. The one or more non-transitory computer-readable media of claim 1, wherein the specific group of CDRs are stored in the data archive store for a predetermined period of time before being purged from the data archive store.

11. The one or more non-transitory computer-readable media of claim 1, wherein the inputted selection parameters are used to filter one or more CDRs from the specific group of CDRs.

12. The one or more non-transitory computer-readable media of claim 1, wherein the generating includes directing the one or more collector applications to retrieve the particular set of CDRs from the data archive store via corresponding CDR location paths to the particular set of CDRs in the data archive store.

13. The one or more non-transitory computer-readable media of claim 12, wherein the re-rating acts further include modifying one or more of the CDR location paths based on one or more inputted data source path modifications.

14. The one or more non-transitory computer-readable media of claim 1, wherein the re-rating acts further include modifying one or more data destination paths used by the one or more CDR distributor applications to send the reprocessed CDRs to the one or more billing systems based on one or more inputted data destination path modifications.

15. A computer-implemented method, comprising:

monitoring, via one or more computing devices, one or more network components of a wireless carrier network;

detecting, via the one or more computing devices, that an anomalous event that is external to the wireless carrier network occurred and negatively impacted a particular set of subscribers;

based on the anomalous event that is external to the wireless carrier network occurring and negatively impacting the particular set of subscribers, determining to modify the service fees charged to the particular set of subscribers;

identifying, via the one or more computing devices, a particular set of CDRs associated with the particular set of subscribers that are negatively impacted by the anomalous event; and

modifying, via the one or more computing devices, the service fees charged to the particular set of subscribers by generating reprocessed CDRs that correspond to the particular set of CDRs for processing by one or more billing systems.

16. The computer-implemented method of claim 15, wherein the modifying further comprises:

importing a business logic that is used to generate call details records (CDRs) for a plurality of subscribers that use wireless telecommunication services provided by a wireless carrier network;

receiving a selection of one or more data collector applications of a plurality of data collector applications that are listed in the business logic, the one or more data collector applications corresponding to a specific group of CDRs, the plurality of data collector applications configured to collect CDRs from one or more servers of the wireless carrier network as the plurality of subscribers use the wireless telecommunication services;

updating one or more configuration parameters in the business logic based on one or more inputted parameter updates;

receiving a selection of one or more CDR distributor applications of a plurality of CDR distributor applications listed in the business logic, the plurality of CDR distributor applications configured to distribute the CDRs to the one or more billing systems of the wireless carrier network;

selecting a particular set of CDRs in the specific group of CDRs that belong to the particular set of subscribers based on one or more inputted selection parameters;

generating reprocessed CDRs that correspond to the particular set of CDRs by directing the one or more collector applications to retrieve the particular set of CDRs from a data archive store of the wireless carrier network and reprocess the particular set of CDRs according to at least the one or more updated configuration parameters in the business logic; and

sending the reprocessed CDRs to the one or more billing systems via the one or more CDR distributor applications to trigger recalculation of service fees charged to the particular set of subscribers based at least on the reprocessed CDRs by the one or more billing systems.

17. The computer-implemented method of claim 16, wherein the modifying further include determining whether the specific group of CDRs is currently archived in the data archive store of the wireless carrier network in addition to being previously sent to the one or more billing systems, and wherein the updating the one or more configuration parameters, the receiving a selection of one or more CDR distributor applications, the selecting the particular set of CDRs, and the generating the reprocessed CDRs, and the sending the reprocessed CDRs are performed when the specific group of CDRs is archived in the data archive store.

18. The computer-implemented method of claim 15, where the detecting includes querying one or more data stores of the wireless carrier network to identify the particular set of subscribers.

19. A computing device, comprising:

one or more processors; and

memory including a plurality of computer-executable components that are executable by the one or more processors to perform a plurality of actions that comprise re-rating actions, the re-rating actions comprising:

importing a business logic that is used to generate call details records (CDRs) for a plurality of subscribers that use wireless telecommunication services provided by a wireless carrier network;

receiving a selection of one or more data collector applications of a plurality of data collector applications that are listed in the business logic, the one or more data collector applications corresponding to a specific group of CDRs, the plurality of data collector applications configured to collect CDRs from one or more servers of the wireless carrier network as the plurality of subscribers use the wireless telecommunication services;

updating one or more configuration parameters in the business logic based on one or more inputted parameter updates;

receiving a selection of one or more CDR distributor applications of a plurality of CDR distributor applications listed in the business logic, the plurality of CDR distributor applications configured to distribute the CDRs to one or more billing systems of the wireless carrier network;

determining that an anomalous event that is external to the wireless carrier network occurred and negatively impact a particular set of subscribers;

based on the anomalous event that is external to the wireless carrier network occurring and negatively impacting the particular set of subscribers, determining to modify the service fees charged to the particular set of subscribers;

selecting a particular set of CDRs in the specific group of CDRs that belong to the particular set of subscribers based on one or more inputted selection parameters;

generating reprocessed CDRs that correspond to the particular set of CDRs by directing the one or more collector applications to retrieve the particular set of CDRs from a data archive store of the wireless carrier network and reprocess the particular set of CDRs according to at least the one or more updated configuration parameters in the business logic; and

sending the reprocessed CDRs to the one or more billing systems via the one or more CDR distributor applications to trigger recalculation of service fees charged to the particular set of subscribers based at least on the reprocessed CDRs by the one or more billing systems.

20. The computing device of claim **19**, further comprising actions that include:

querying one or more data stores of the wireless carrier network to identify the particular set of subscribers; and

performing the re-rating actions to modify the service fees charged to the particular set of subscribers.

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